Collaborative Discussion 3: Deep Learning

This discussion will last for 3 weeks, starting in Unit 9 and ending in Unit 11. Ensure you include appropriate citations and references in all your posts.

Discussion Topic: Deep Learning

The advent of new technologies supported by Deep Learning models mean that it is now possible to generate 'new' content, for example, Dall-E AI to generate images or ChatGPT to create prose.

Do you think that these new technologies offer any ethical issues that should be considered, and if not, why not?

Learning Outcomes

- The knowledge and skills required to develop, deploy and evaluate the tools and techniques of intelligent systems to solve real-world problems.

Assignment Guidance

- Your initial posting should respond to the question and be at least 200 words long. This should be labelled as '**Initial Post**'
- You will then respond to **at least 2 of your peers' posts** in Unit 10 (each labelled as '**Peer Response**'). To guide your responses, look at the guidelines for the peer review process on the **Department's homepage**. Focus on the possible measures that could have been put in place in order to prevent the incidents highlighted by your peers. Please try to limit your response posts to **200-300 words maximum**, so that others may be encouraged to reflect on, and respond to your ideas.
- In Unit 11, you should provide a summary post based on your initial post, the feedback from your peers and the content of the three units. Please label this as '**Summary Post**'. It should be **300 words**.
- Referencing: When you have referred to other authors thoughts, ideas and opinions in your posts, you must reference the author as you would in an academic assignment using the UoEO Harvard reference style.

- You will be not be assessed on your contribution to this forum throughout the module, but the tutor will post group feedback via a module announcement.
- This activity forms a component of your e-portfolio which you will submit in Unit 12. All e-portfolio activities are intended to demonstrate your ability and strengths through evidence and reflection.

Initial Post

by Yasmin Almeida Ferrera - Monday, 28 September 2026, 9:29 AM
Number of replies: 3

The development of generative Deep Learning models such as DALL-E and ChatGPT offers significant opportunities, but I believe their ethical implications should be considered as seriously as their technical capabilities. In particular, the ability to generate convincing content creates concerns around trust, accountability and academic integrity.

This is something I have considered in my own MSc project, ScholarPlan, an LLM-powered academic research planning agent. The system is designed to retrieve academic sources, generate source-linked claims and verify them. However, this highlights an important limitation of generative AI: **hallucination**. Research by Walters and Wilder (2023) found that ChatGPT could generate fabricated bibliographic references, demonstrating that apparently credible academic output cannot automatically be considered reliable.

For me, this makes human oversight essential. ScholarPlan therefore includes verification and an execution trace, allowing users to examine where information comes from rather than simply trusting the generated output. This reflects UNESCO's emphasis on transparency, accountability and human oversight.

NIST (2024) similarly identifies “confabulation” as a significant generative-AI risk, particularly when users act on false information or citations.

Personally, I believe generative AI should not be rejected because of these risks, but its development should be accompanied by mechanisms that make AI verifiable, transparent and accountable.

If an AI system produces an incorrect academic claim despite verification mechanisms, where should responsibility ultimately lie: with the developer, the user, or both?

References

NIST (2024) *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1. Gaithersburg, MD: National Institute of Standards and Technology. Available at: **https://doi.org/10.6028/NIST.AI.600-1** (Accessed: 28 September 2026).

UNESCO (2021) *Recommendation on the Ethics of Artificial Intelligence*. Paris: UNESCO. Available at: **https://www.unesco.org/en/artificial-intelligence/recommendation-ethics** (Accessed: 28 September 2026).

Walters, W.H. and Wilder, E.I. (2023) 'Fabrication and errors in the bibliographic citations generated by ChatGPT', *Scientific Reports*, 13, 14045. Available at: **https://doi.org/10.1038/s41598-023-41032-5** (Accessed: 28 September 2026).

Peer Response

by Panagiotis Monogios - Monday, 28 September 2026, 12:12 PM

Yasmin, I agree that hallucination is particularly dangerous in academic systems because generated claims and references can appear credible even when they are incorrect. Walters and Wilder (2023) demonstrate this clearly through fabricated and erroneous bibliographic citations. Your use of retrieval, verification and an execution trace in ScholarPlan therefore seems like a sensible way of reducing blind reliance on generated output.

However, I would argue that verification mechanisms should themselves be treated as fallible. If one model generates a claim and another automated component checks it, both may still rely on incomplete retrieval, incorrect metadata or similar underlying assumptions. Preventive measures could therefore include checking claims against primary sources, requiring source identifiers such as DOI values where possible, recording confidence or verification status, and escalating uncertain claims for human review rather than presenting them as verified.

On your question about responsibility, I do not think it should fall entirely on either the developer or the user. Responsibility should depend partly on control and foreseeable risk. Developers are responsible for designing reasonable safeguards, communicating limitations and avoiding interfaces that create unjustified confidence. Users also have responsibilities when they rely on outputs in contexts where verification is expected.

UNESCO (2021) and NIST (2024) both emphasise accountability and human oversight rather than assuming that responsibility disappears once an AI system is deployed.

For an academic research agent, I would therefore favour shared but clearly defined responsibility: the system should provide traceable evidence, while the researcher remains responsible for validating claims before relying on them.

Would you allow ScholarPlan to mark a source as “verified” automatically, or would final verification always require human confirmation?

References

National Institute of Standards and Technology (NIST) (2024) *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1. Gaithersburg, MD: NIST.

UNESCO (2021) *Recommendation on the Ethics of Artificial Intelligence*. Paris: UNESCO.

Walters, W.H. and Wilder, E.I. (2023) ‘Fabrication and errors in the bibliographic citations generated by ChatGPT’, *Scientific Reports*, 13, 14045.

Peer Response

by Isaac Shalaan - Friday, 2 October 2026, 12:49 AM

Dear Yasmin Almeida Ferrera,

Your ScholarPlan example highlights an important ethical distinction. Making an academic claim traceable does not necessarily make it accurate. I agree that verification is essential, but would extend this to checking whether each cited passage supports the associated claim.

Gao *et al.* (2023) distinguish citation quality from answer correctness, demonstrating that generated answers can lack complete evidential support. For ScholarPlan, I would recommend separate checks for bibliographic accuracy and agreement between claims and sources, with supporting passages displayed beside claims. This could help users identify genuine references that have been misinterpreted.

A second safeguard would be allowing the system to withhold unsupported answers. Kalai *et al.* (2026) show that accuracy-based evaluation can incentivise guessing rather than acknowledging uncertainty. ScholarPlan could therefore be tested using deliberately unanswerable questions, with unsupported assertions penalised more heavily than appropriate abstention. Farquhar *et al.* (2024) demonstrate that semantic entropy, which measures variation in meaning across generated answers, can detect some hallucinations. However, it does not address consistently incorrect outputs, so it should supplement source verification.

Human oversight also warrants evaluation. Vaccaro, Almaatouq and Malone's (2024) meta-analysis found that combinations of humans and AI, on average, underperformed the better of humans or AI alone. Although not specific to academic agents, this cautions against assuming that adding a reviewer guarantees reliability. User testing should establish whether ScholarPlan's execution trace helps researchers detect errors.

Regarding responsibility, I would argue for shared but differentiated ethical accountability. Developers should test foreseeable failures, communicate limitations and enable correction, while users should examine evidence before incorporating claims into academic work. Responsibility should reflect each party's control over the failure.

ScholarPlan's success should therefore be assessed partly by how reliably it withholds claims when supporting evidence is insufficient.

References:

Farquhar, S., Kossen, J., Kuhn, L. and Gal, Y. (2024) 'Detecting hallucinations in large language models using semantic entropy', *Nature*, 630, pp. 625–630. Available at: **https://doi.org/10.1038/s41586-024-07421-0** (Accessed: 2 October 2026).

Gao, T., Yen, H., Yu, J. and Chen, D. (2023) 'Enabling large language models to generate text with citations', in Bouamor, H., Pino, J. and Bali, K. (eds.) *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Singapore: Association for Computational Linguistics, pp. 6465–6488. Available at: **https://aclanthology.org/2023.emnlp-main.398/** (Accessed: 2 October 2026).

Kalai, A.T., Nachum, O., Vempala, S.S. and Zhang, E. (2026) 'Evaluating large language models for accuracy incentivizes hallucinations', *Nature*, 653, pp. 1047–1051. Available at: **https://doi.org/10.1038/s41586-026-10549-w** (Accessed: 2 October 2026).

Vaccaro, M., Almaatouq, A. and Malone, T. (2024) 'When combinations of humans and AI are useful: A systematic review and meta-analysis', *Nature Human Behaviour*, 8, pp.

2293–2303. Available at: **https://doi.org/10.1038/s41562-024-02024-1** (Accessed: 2 October 2026).

Re: Initial Post

by Yasmin Almeida Ferrera - Monday, 12 October 2026, 11:11 AM

Summary Post

Generative deep learning models such as ChatGPT offer valuable opportunities for research and information gathering, but their ethical implications must be considered alongside their technical capabilities. In my initial post, I discussed hallucinations, academic integrity and the importance of transparency, using my MSc project, ScholarPlan, as an example. Walters and Wilder (2023) demonstrated that ChatGPT can produce fabricated bibliographic references, highlighting why AI-generated information cannot automatically be trusted.

The feedback from my peers helped me recognise that traceability and verification do not necessarily guarantee accuracy. Panagiotis emphasised that automated verification can also fail and suggested checking primary sources, recording verification status and involving humans when uncertainty remains. Isaac extended this argument by highlighting the importance of checking whether cited passages actually support generated claims. He also suggested that ScholarPlan should be able to withhold unsupported answers rather than generate potentially misleading information.

These perspectives changed how I think about evaluating AI systems. Accuracy alone is insufficient; reliability, evidential support and the ability to acknowledge uncertainty are equally important. The wider content of the three units reinforced the need to consider transparency, accountability, privacy and the social consequences of AI deployment. As UNESCO (2021) and NIST (2024) emphasise, responsible AI requires appropriate safeguards and human oversight.

Regarding responsibility, I agree with both peers that accountability should be shared but clearly differentiated. Developers should design safeguards, test foreseeable failures and communicate limitations, while users must critically assess evidence before relying on AI-generated information. However, human oversight should not be assumed to eliminate errors automatically; it must also be evaluated.

Overall, this discussion has strengthened my view that generative AI should support rather than replace human judgement. For ScholarPlan, success should involve not only retrieving sources and generating claims, but also demonstrating evidential support, identifying uncertainty and withholding unsupported conclusions. Responsible AI development requires continuous evaluation, transparency and accountability to ensure

that technical innovation benefits users without creating unjustified confidence in its outputs.

References

National Institute of Standards and Technology (NIST) (2024) *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1. Gaithersburg, MD: National Institute of Standards and Technology. Available at: **https://doi.org/10.6028/NIST.AI.600-1** (Accessed: 28 September 2026).

UNESCO (2021) *Recommendation on the Ethics of Artificial Intelligence*. Paris: UNESCO. Available at: **https://www.unesco.org/en/artificial-intelligence/recommendation-ethics** (Accessed: 28 September 2026).

Walters, W.H. and Wilder, E.I. (2023) 'Fabrication and errors in the bibliographic citations generated by ChatGPT', *Scientific Reports*, 13, article 14045. Available at: **https://doi.org/10.1038/s41598-023-41032-5** (Accessed: 28 September 2026).